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Better Spotify / Sample Browser

Granular Audio Passage Discovery for Independent Music Producers

Project Report — Midterm 1

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Abstract

The project **"Better Spotify / Sample Browser"** (hereinafter referred to as **Sample Browser**) was developed by students Bogdan Glavan, Andrei Simon, Nikita Creciun, Leonid Tiscenco, and Mihai-Constantin Sîrghii, under the mentorship of Prof. Elena Graur at the Technical University of Moldova, Faculty of Computers, Informatics and Microelectronics. This report encapsulates the initial Conceptual Design of a Software Application (PCAS) stage, covering problem definition, domain analysis, methodological evolution, solution architecture, target audience validation, and comparative benchmarking.

Keywords: Better Spotify, Sample Browser, Audio Sampling, Music Information Retrieval (MIR), Granular Passage Discovery, Music Streaming Architectures, Creator Workflow, Problem-Based Learning.

The primary objective of **Sample Browser** is to eliminate the persistent workflow friction experienced by independent music producers during sample crate-digging across streaming catalogs. Current digital streaming platforms index musical compositions exclusively at the whole-track metadata level, ignoring sub-track time boundaries. Consequently, producers seeking specific localized musical textures—such as isolated 4-bar rhythm breaks or distinct instrumental passages—must scrub back and forth manually across long audio files, disrupting their creative focus. To solve this, **Sample Browser** provides a specialized discovery platform that interfaces with embedded streaming audio via the YouTube IFrame API to segment, tag, and recommend localized audio passages (4–30 seconds). By coupling a structured Two-Tier Tagging System (objective structural presets and contextual descriptors) with direct timestamp seeking and automatic loop auditioning, the system transforms unstructured listening into a targeted creator utility. Through a rigorous Problem-Based Learning methodology, this report documents the team's pivot from an unfeasible consumer streaming overlay, establishes four verifiable validation hypotheses (H1–H4), benchmarks market alternatives, and provides the architectural foundation for subsequent systems modeling and empirical user validation.

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1 Introduction

Streaming catalogs currently host over 250 million recordings, with 100,000 new tracks uploaded every single day [14]. While casual listeners enjoy endless background music, independent music producers hunting for samples face an exhausting search process. Audio sampling underpins roughly 17% of Billboard Hot 100 hits [37]. Yet commercial streaming platforms treat every song as an undifferentiated, continuous audio file. Producers rarely sample entire songs. Instead, they search for localized 4-bar drum breaks, clean keyboard progressions, or solo intros without vocals.

Our project, **"Better Spotify / Sample Browser"** (hereinafter referred to as **Sample Browser**), tackles this bottleneck directly. More than half of active music makers work regular jobs outside the music industry [13]. They produce music during limited spare hours alongside their day jobs. Spending entire evenings dragging playheads across long recordings quickly drains creative energy. **Sample Browser** resolves this issue by indexing, tagging, and recommending music at the passage level (4 to 30 seconds), jumping creators straight to the cue points they need.

The core objective of this research is to design and validate the conceptual architecture of **Sample Browser**. The platform interfaces with public embedded audio via the YouTube IFrame API to deliver timestamped passage recommendations, avoiding media hosting costs and copyright risks. Guided by the Problem-Based Learning (PBL) framework, we grounded our engineering decisions in Music Information Retrieval (MIR) literature, creator surveys, and platform policy audits—specifically Spotify's 2024–2026 developer restrictions on audio feature endpoints [35, 36].

The following chapters map out our research trajectory and conceptual design. Chapter 2 details the problem analysis, documents our team's early methodological errors and iterative pivot, presents our solution architecture, establishes target personas with testable validation hypotheses (H1–H4), and benchmarks market alternatives alongside a strategic SWOT matrix. Chapter 3 concludes the report with our roadmap for Midterm 2.

2 Midterm 1

The first milestone of this project focuses on identifying, scoping, and analyzing an operational problem within modern music streaming and digital production workflows. Guided by the Problem-Based Learning (PBL) paradigm, this chapter documents the analytical foundation of our project: crystallizing the problem description, performing an industry-wide domain analysis, reflecting on early methodological misconceptions and iterative pivots, defining the conceptual solution architecture for **Sample Browser**, segmenting the target audience with verifiable hypotheses, and performing a comparative benchmarking against market alternatives.

2.1 Problem Description and Problem Analysis

Problem Description

Sampling and reference-based production fundamentally rely on discovering specific musical textures, localized breaks, and unlayered passages within existing recordings. However, modern digital audio distribution channels treat musical works as single, undivided files, creating persistent friction for creators.

Target Demographic and Creative Context

The demographic experiencing this problem comprises independent music producers, beatmakers, and sound designers working in sample-intensive and electronic genres (e.g., hip-hop, electronic, neo-soul, lofi, and experimental music). According to Tracklib's 2024 Producer Survey ($N = 1,107$), aspiring producers aiming to build a full-time career constitute 58% of creators, while hobbyists (25%) and beginners (8%) make up the majority of the remainder; full-time professionals represent only 10% of the creator population [38].

Importantly, findings from the UK Musicians' Census ($N \approx 6,000$) reveal that 53% of active music makers must sustain their living through employment outside the music industry [13, 4]. Independent creators therefore operate with limited spare time alongside full-time jobs. Hours spent on repetitive, non-creative manual search tasks directly diminish the limited time available for actual sound design, arrangement, and composition.

The Dual Bottleneck of Sample Discovery

When an independent producer auditions full-length recordings across public streaming platforms to find sample material, they encounter two structural bottlenecks, as visualized in Figure 1:

1. **Sub-Track Structural Blindness (The Whole-Track Metadata Bottleneck):** Streaming platforms (such as YouTube) and discovery overlays (such as Samplette.io) index recordings exclusively at the whole-track metadata level (Title, Artist, Global Genre, Year, overall BPM). Yet a music producer rarely samples an entire song; they search for a localized, **structurally isolated passage**—such as an

unlayered 4-bar intro, a clean drum break, or a solo electric piano progression without vocals or heavy percussion bleeding over it. Without sub-track structural indexing, producers cannot see where usable moments occur, forcing them to manually click and scrub the playhead across dozens of 3-to-6 minute recordings just to check if an isolated section exists.

2. **Ambiguity of Unconstrained Text Queries and Bookmarking Friction:** When producers attempt to search for specific sounds using arbitrary natural-language descriptions, subjective mood or texture keywords yield inconsistent results because listeners interpret free-form adjectives differently. In Music Information Retrieval (MIR) literature, Andersen and Knees [17] observed that musicians rely on subjective metaphors (”bright”, ”gritty”, ”hollow”) and visual associations, while Zhang, Zhang, and Duan [42] and Kreković [12] demonstrated that complex mixed audio textures lack semantically agreed-upon free-text descriptors. Furthermore, as Flexer [50] demonstrated, low inter-rater agreement on perceptual music labels undermines arbitrary text tagging without auditory grounding. Coupled with the fact that standard browser bookmarks and YouTube playlists only save full-track URLs, creators face severe friction re-identifying specific 10-second passages and timestamps weeks later.

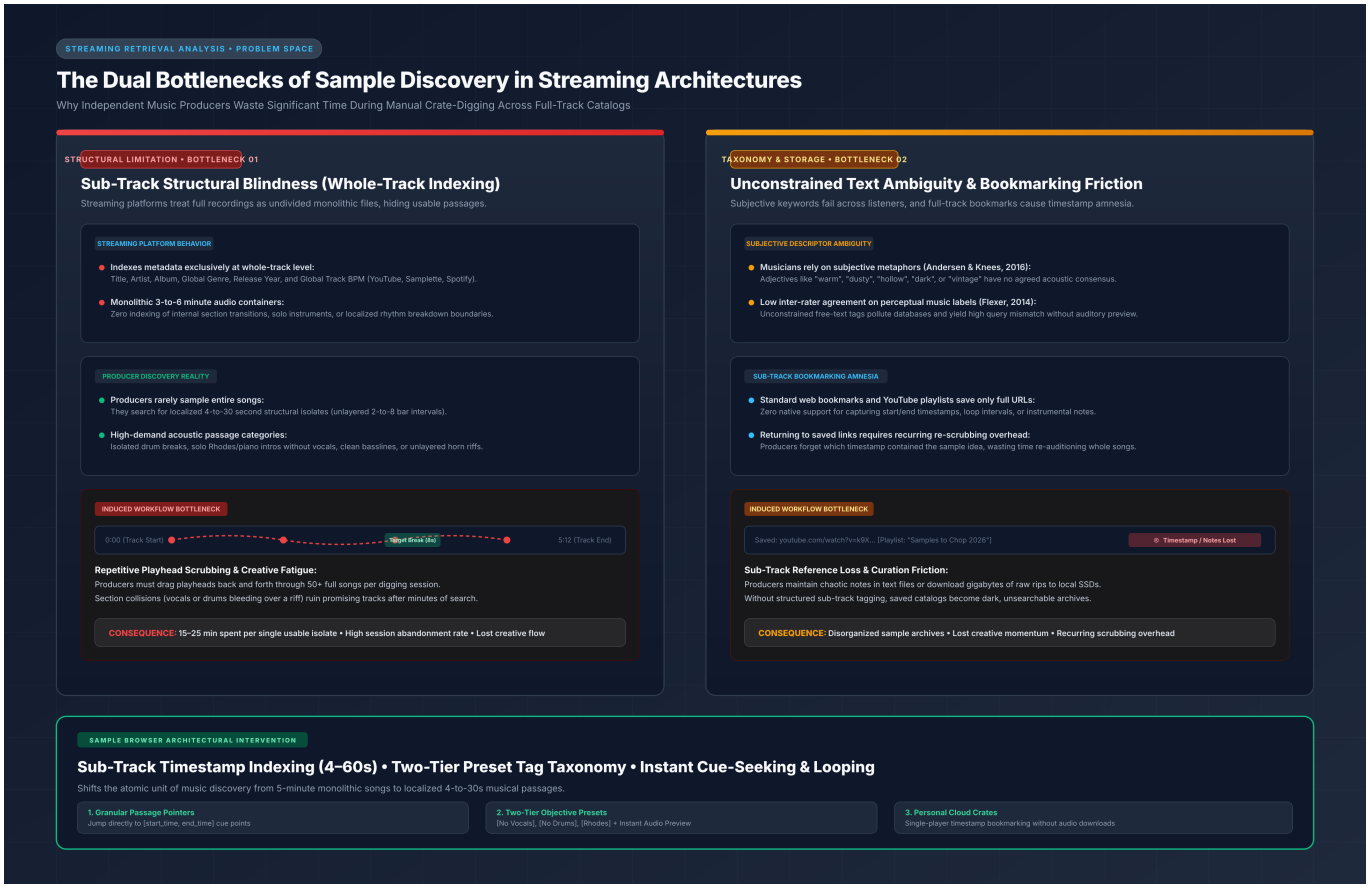


Figure 1 - The Dual Bottlenecks of Sample Discovery in Streaming Architectures.

This repetitive audition-and-scrub cycle induces scrubbing fatigue and destroys the producer’s creative flow state, a phenomenon analyzed by Nash and Blackwell [3]. While traditional ”crate-digging”

in physical record stores historically celebrated serendipitous discovery [16, 41], navigating flat, multi-million-track digital databases without temporal indexing turns discovery into an exhausting manual audit.

Stakeholder Landscape

- **Primary Stakeholders (Independent Music Producers):** Benefit from direct, friction-free passage discovery, reducing audition time and preserving cognitive momentum during the creative process.
- **Secondary Stakeholders (Original Rights Holders and Catalog Artists):** Back-catalog recordings and obscure archival tracks gain renewed relevance, engagement, and potential licensing revenue when surfaced to active samplers.
- **Tertiary Stakeholders (Streaming Platforms and Audio Services):** Benefit from higher creator retention, transforming passive consumer players into active production and curation hubs.

Domain Relevance and Scale

Sampling is not a fringe artistic practice; it is foundational to modern music production. Statistical industry analyses indicate that approximately 17% of Billboard Hot 100 charting singles incorporate sampled material [37]. At the same time, Luminate and IFPI report that global audio streaming catalogs expanded past 253 million tracks, with over 106,000 new recordings uploaded daily [14, 20]. Without sub-track indexing, this rapid expansion makes locating distinctive passages increasingly difficult, burying valuable musical recordings behind rigid metadata walls.

Domain Analysis

Platform Recommendation Paradigms and Algorithmic Concentration

Modern streaming platforms categorize music at wildly differing levels of granularity, primarily optimized for passive consumption rather than creative exploration. Spotify—commanding 31.4% of global streaming subscriptions [19]—relies on black-box recommendation algorithms trained on long-term user telemetry (skips, listening duration, playlist adds) [17], drawing on collaborative filtering frameworks popularized by platforms such as Last.fm [60]. To address discovery complaints, Spotify introduced conversational AI tools, including its AI DJ (launched in beta in 2023 and updated with voice requests in 2025) [52] and AI Playlist (prompt-to-playlist generation in 2024) [54]. However, independent evaluations highlight clear limitations: user audits identify recurrent song loops and genre homogenization [53]. Recommenders trained on historical consumption inherently narrow catalog exposure: simulation studies prove algorithmic confounding increases homogeneity while decreasing user utility [2, 1], with empirical consumer studies showing that algorithmic nudging restricts diversity [23], and streaming audits confirming that automated recommendations heavily concentrate within a small fraction of catalog tracks [15, 63].

Recent features such as Spotify's SongDNA (powered by WhoSampled) surface documented historical sample relationships and interpolations [31, 33, 34]. However, SongDNA is strictly retrospective: it exposes known connections between existing hits rather than facilitating open, acoustic-similarity search

across unlinked catalog recordings.

In contrast to Spotify's behavioral learning, alternative streaming platforms emphasize immediate user controls. Deezer offers explicit mood-based session filtering through Flow Moods (e.g., Chill, Focus, Melancholy) [51]. Yandex Music's "My Wave" enables listeners to restrict playback to instrumental tracks ("without words") or steer recommendations toward unfamiliar artists via an explicit "Unfamiliar" mode [61], while VK Music allows compound multi-parameter filtering [62]. While these immediate controls mitigate common listener frictions, they remain restricted to whole-track playback and offer no visibility into localized sub-track musical events.

Divergent Information-Seeking Objectives: Casual Listeners vs. Music Producers

Information-seeking behavior in musical domains differs fundamentally based on user goals. In Information Science literature, Laplante and Downie [47] categorize listener behavior into utilitarian needs (searching for a known track or artist) and hedonic discovery (exploring moods, serendipity, and emotional resonance), supported by diary studies on everyday novel song encounters [49, 64]. Mainstream listeners accept whole-track recommendations because their objective is continuous audio accompaniment.

By comparison, music producers exhibit specialized, analytical retrieval workflows. As demonstrated by Demetriou et al. [48], producers search for sound as raw structural material rather than finished compositions. They seek specific acoustic isolates: a 4-bar drum break, an unlayered electric piano progression, or an isolated vocal timbre suitable for rhythmic manipulation. Furthermore, physical crate-digging historically carried profound social capital within hip-hop and electronic music cultures [16, 41]. Producers prize obscurity—discovering recordings that few others have heard or repurposed. In online ecosystems, producers even co-opt search behavior through conventions such as "[Artist] Type Beats", functioning as informal search engine optimization to locate specific production aesthetics [58].

Catalog Saturation, The Generative AI Music Surge, and Audio LLM Limits

The challenge of locating obscure, high-quality sample material is compounded by the unprecedented saturation of digital catalogs. In July 2026, streaming platform Deezer reported that AI-generated tracks accounted for over 50% of its daily uploads, representing approximately 90,000 synthetic tracks uploaded each day [43]. This synthetic flood drowns out authentic archival recordings, making manual digging through new releases increasingly frustrating.

At the same time, music creators exhibit notable resistance toward fully autonomous generative music. In Tracklib's 2024 Producer Survey ($N = 1,107$), 75% of producers reported that they do not use generative AI, with 82.2% citing artistic reasons as their primary objection [56]. This resistance intensified in Tracklib's 2025 survey of 1,734 music creators: an overwhelming 82% opposed generating complete songs from text prompts, while enthusiastically supporting assistive, non-generative tools such as stem separation and intelligent search [44]. As producers expressed in field interviews with Andersen and Knees [17], cre-

ators welcome algorithmic assistance only when they retain ultimate authorship: ”I would like the machine to sit in the chair, as long as I get to decide when it gets out.”

Furthermore, modern Multimodal Audio-Language Models (audio-LLMs) struggle to evaluate raw musical sound accurately. In a comprehensive 2025 benchmark evaluating 10,000 audio clips across speech, environmental sound, and music, the leading foundation models achieved only 60% question-answering accuracy [46]. Similarly, the MuChoMusic evaluation framework presented at ISMIR 2024 revealed that audio-language models over-rely on linguistic cues within the text prompt rather than truly extracting acoustic features from the audio signal [45]. Because text queries struggle to capture nuanced sonic textures [42, 12], fully automated, text-prompted discovery remains unreliable. This underscores the necessity of human-in-the-loop, structured acoustic tagging.

Current Producer Workflows and The Granularity Divide

Producers currently navigate this fragmented landscape using disconnected tools that expose a fundamental structural divide:

- **Snippet Libraries (Splice, Looperman):** Splice—surpassing 350 million downloads in 2024 [21]—indexes short, pre-cut samples with timbre-matching tools (*Similar Sounds*, *Search with Sound*, *Describe a Sound*) [27, 26, 28, 29]. However, Splice operates strictly within a closed, proprietary sound pack catalog. Millions of producers draw from the identical pool of sounds, creating widespread ”sample fatigue.”
- **Pre-Cleared Sampling Catalogs (Tracklib):** Provides full master recordings cleared for legal sampling [37, 39, 24]. Tracklib incorporates a bar-looping tool that slices playback into 1, 2, 4, or 8 bars and provides stem isolation [55, 57]. However, Tracklib categorizes records strictly by artist, genre, and BPM at the whole-track level; its looping tool functions only *after* the producer has already found and manually auditioned the song.
- **Randomized Streaming Digging (Samplette.io):** Interfaces with YouTube to deliver random obscure tracks filtered by style, BPM, and a custom view-count ceiling (e.g., serving only videos with under 10,000 views) [25, 22, 59]. While Samplette honors the crate-digger’s desire for obscurity, it indexes at the whole-track level: users must still scrub manually through every 4-minute video.
- **Enterprise Audio Intelligence (Cyanite.ai):** Offers state-of-the-art acoustic similarity search and multi-label classification via B2B APIs [5, 6, 7, 8, 10, 11]. However, Cyanite is tailored for sync-licensing agencies and catalog publishers rather than individual beatmakers, requiring enterprise contracts and capped at 30-second Spotify preview snippets [9].

This landscape reveals a critical market void: snippet libraries describe isolated sounds in closed databases, while streaming platforms describe whole songs with coarse metadata. No platform describes, indexes, or recommends the internal, sub-track sections of published musical compositions.

Platform API Constraints and Legal Realities

A critical dimension of domain analysis is the technical feasibility of third-party audio analysis. In November 2024, Spotify initiated sweeping restrictions on its Web API, formally deprecating public access to the `Recommendations`, `Audio Features`, and `Audio Analysis` endpoints for all newly registered applications [35]. In February 2026, Spotify further tightened development guidelines by restricting development mode to a hard cap of 5 authorized test users, mandating Spotify Premium for playback, and enforcing developer policies that explicitly prohibit ingesting Spotify audio streams into external machine learning models [36, 30].

Similarly, YouTube's Developer Terms prohibit stripping audio, background playback, or bypassing the graphical player interface. Consequently, any viable third-party creator tool cannot rely on unauthorized audio downloading or restricted Spotify endpoints; it must operate strictly through compliant embedded streaming architectures (such as the YouTube IFrame API).

Project Evolution, Failures, and Mistake Analysis

In accordance with PBL principles, documenting false starts, methodological misconceptions, and iterative course corrections is essential for establishing sound engineering judgment. The project originally began under the working title "Better Spotify" as an alternative consumer streaming shell intended to counter algorithmic bubbles through gesture-controlled queue flows and tag filters (Figure 2) [40].

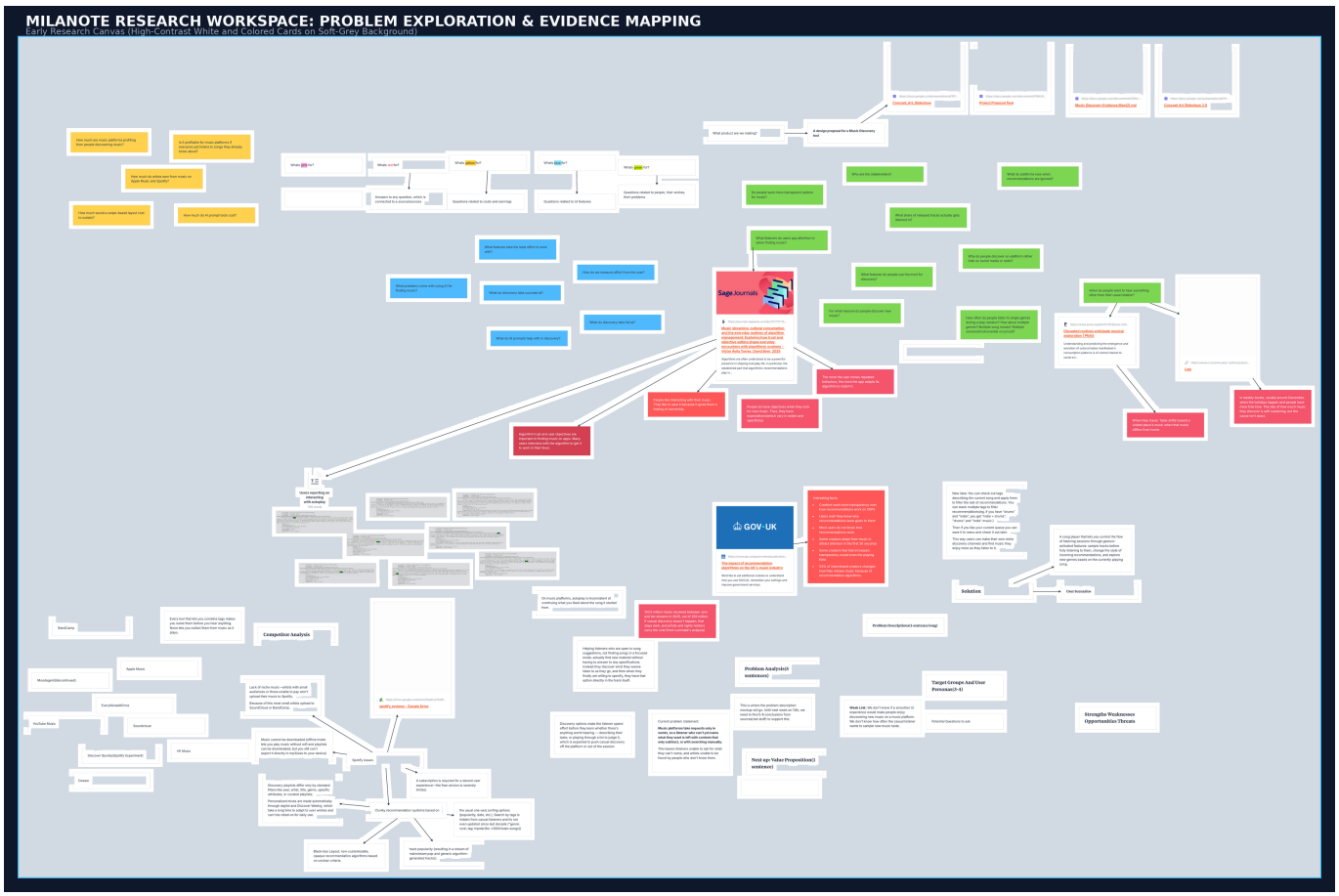


Figure 2 - Early Research Evidence Map and Problem Brainstorming Canvas on Milanote.

During iterative critique and mentor evaluation, this initial vision was critically dismantled due to three systemic methodological flaws:

1. **Misinterpreting Vocal User Complaints:** Negative app store reviews over-represented a vocal minority. In reality, over 80% of consumers are passive, lean-back listeners who easily bypass unappealing tracks by hitting ”Next” with negligible cognitive effort.
2. **Overlooking Platform Governance and API Lockdowns:** The concept assumed external control over proprietary streaming queues. Spotify’s updated 2024–2026 Developer Policy explicitly deprecated audio analysis endpoints and enforced strict playback boundaries, making a custom consumer playback shell legally and technically non-viable [35, 36, 30].
3. **Targeting Casual Inconvenience over Mission-Critical Friction:** True product viability requires high stakes. While casual listeners face negligible switching friction, music producers operate under severe time constraints, where manual searching actively damages creative output.

Recognizing these systemic barriers, the team executed an intentional pivot to **”Better Spotify / Sample Browser”** (hereinafter referred to as **Sample Browser**). Shifting the target demographic from passive consumers to sample-based creators transformed the product from an unfeasible playback shell into an active creator utility, indexing music at the passage level while operating strictly within compliant

embedded playback boundaries.

Formal Problem Statement

"Independent music producers digging for samples across public streaming catalogs (such as YouTube and Samplette) waste significant time manually scrubbing through full-length recordings, because existing platforms index metadata exclusively at the whole-track level and provide no visibility into where short, structurally isolated passages (such as vocal-free instrumental sections or clean drum breaks) occur within a song."

2.2 Solution Proposal

To solve the identified problem, we propose **Sample Browser**, a web-based sub-track discovery and sample-bookmarking platform that shifts the atomic unit of music search from the entire composition to the **sub-track musical passage**.

System Architecture and Core Mechanics

Sample Browser functions as a zero-media-hosting, pointer-based discovery workspace built directly on top of public streaming video via the official **YouTube IFrame Player API**. To remain 100% compliant with Section III.E of the YouTube Developer Policies and completely eliminate bandwidth and hosting costs:

1. The platform **does not download, host, extract, cache, or algorithmically process raw audio streams** on backend servers.
2. The PostgreSQL database stores **only lightweight metadata pointers**: YouTube Video IDs, sub-track segment timestamps (`start_time`, `end_time`, constrained between 4 and 60 seconds), and structured passage tags submitted and curated by users.
3. All audio playback and auditioning execute strictly client-side inside the official embedded YouTube player using synchronized timestamp seeking (`player.seekTo(start_time, true)`) and automated interval boundary looping.

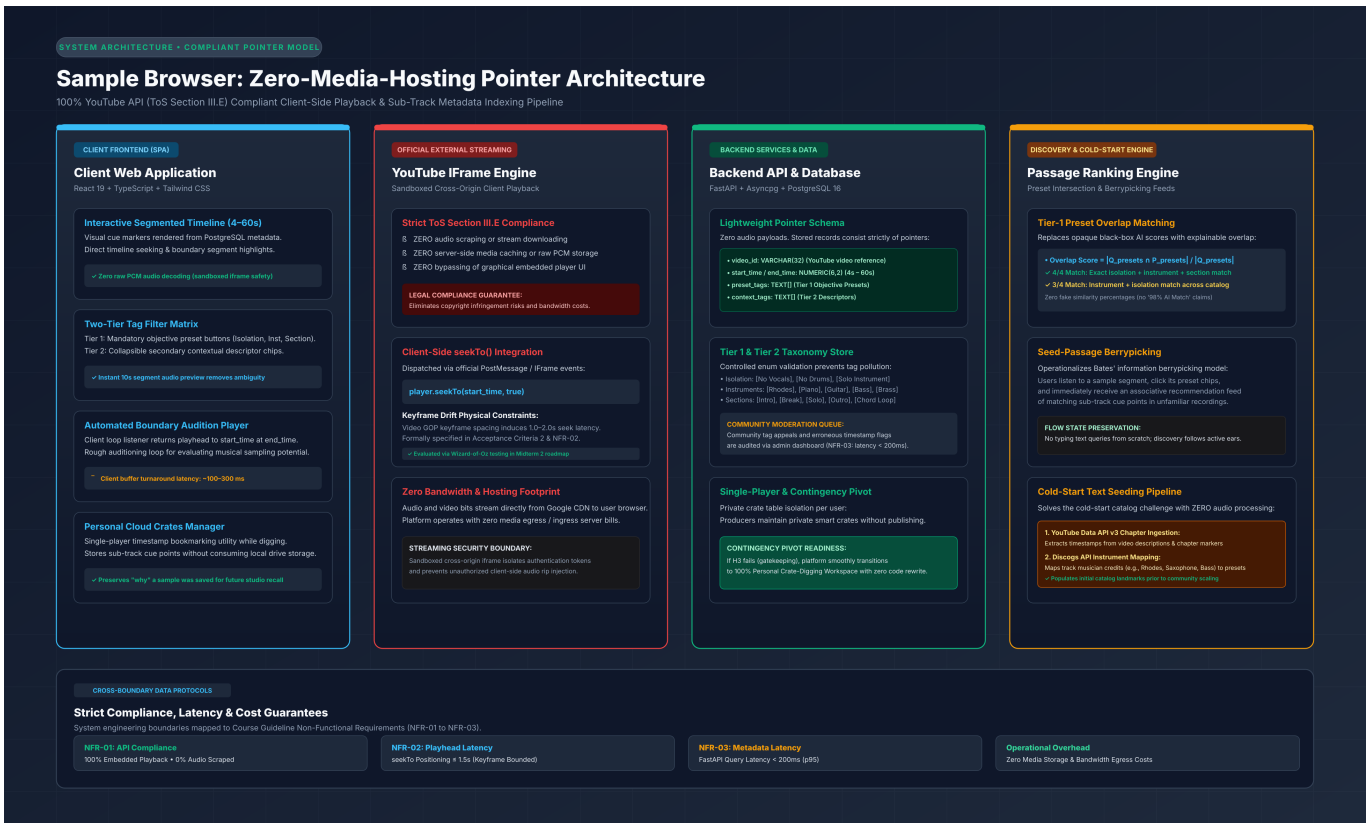


Figure 3 - Sample Browser Pointer-Based Architecture and YouTube IFrame Interaction Pipeline.

1. Interactive Timeline and Segmented Visualizer

When a track is loaded into Sample Browser, the playback timeline displays highlighted segment markers (e.g., `0:00--0:12 [Intro | Rhodes | No Drums]`) retrieved from our PostgreSQL metadata store. Because embedded YouTube players execute inside a sandboxed cross-origin `<iframe>` that denies access to raw PCM audio buffers via the Web Audio API, the client UI does not attempt real-time audio decoding. Instead, the interface visualizes pre-indexed segment cue points delivered by the backend. Clicking a highlighted marker directs the embedded player to jump instantly to `start_time` and automatically loop playback upon reaching `end_time`, allowing producers to audition the isolated passage in rhythm before deciding to chop or sample.

2. Two-Tier Tagging Taxonomy

To resolve the vocabulary gap and prevent database pollution without requiring unfeasible AI audio decoding, Sample Browser replaces arbitrary free-text input with a structured **Two-Tier Tagging System**:

1. **Tier 1 — Primary Preset Objective Tags (Mandatory, High Search Priority):** Selected in 2–3 clicks from a strictly controlled, pre-written taxonomy describing verifiable musical facts about the timestamped segment:
   - *Isolation / Layering Status:* `[No Vocals]`, `[No Drums / Melodic Only]`, `[Isolated Drum Break]`, `[Solo Instrument]`, `[Vocal Phrase / Acapella]`.

- *Instrumentation Present:* `[Electric Piano / Rhodes]`, `[Acoustic Piano]`, `[Guitar]`, `[Bass]`, `[Brass / Sax]`, `[Flute / Winds]`, `[Strings]`, `[Synth / Pad]`, `[Percussion / Drums]`.
- *Structural Section / Function:* `[Intro]`, `[Break / Bridge]`, `[Solo Section]`, `[Outro]`, `[Chord Loop]`, `[One-Shot / Hit]`.

2. **Tier 2 — Secondary Contextual Tags (Optional, Lower Priority, Moderated):** Curated descriptor chips for style and texture inspired by RateYourMusic and Splice categories (e.g., `[Lo-Fi / Dusty]`, `[Syncopated / Swing]`, `[Jazz-Fusion]`, `[Dark / Minor]`). Secondary tags are visually collapsed so they never clutter the primary interface. Crucially, because every tag in Sample Browser is paired with **instant audio playback of the exact timestamped segment**, any subjective ambiguity is immediately resolved by auditioning the 10-second loop in one click.

3. Seed-Passage ("Berrypicking") Navigation and Personal Cloud Crates

Figure 4 contrasts the traditional crate-digging scrubbing cycle with the accelerated Sample Browser workflow. To optimize creator discovery, the platform introduces two interconnected mechanics:

- **Seed-Passage ("Berrypicking") Recommendations:** Rather than forcing creators to invent text search queries from scratch, a producer listening to a highlighted segment can select the tags present on that segment (e.g., `[Guitar] + [No Vocals] + [Break]`) and immediately receive a recommendation feed of specific segments inside other songs sharing those exact preset tags—jumping straight to the matching timestamp in the next track, operationalizing Bates' information berrypicking model [18].
- **Personal Cloud Crates:** While digging across YouTube, producers can mark `Start` and `End` timestamps on any track, attach preset tags, and save the segment into their private Personal Cloud Crate. This eliminates bookmarking chaos and saves SSD disk space by storing lightweight metadata pointers rather than gigabytes of unorganized audio rips.

Cold-Start Seeding and Asynchronous Curation Value

A central architectural concern in human-curated systems is the bootstrapping and scrubbing paradox: if automated audio processing is avoided, does User A still have to scrub through a 5-minute track to tag it for User B, and how does the platform deliver value when the database is initially sparse? Sample Browser addresses this through a three-layer strategy operating without any server-side audio decoding:

1. **Automated Text-Metadata Seeding (Zero Audio Processing):** To populate the initial catalog before user contributions scale, the ingestion pipeline extracts existing structured timestamps from YouTube video descriptions and chapter markers via the official YouTube Data API v3, cross-referencing them against track-level instrument credits from the public Discogs API (e.g., automatically identifying recordings credited with `Fender Rhodes`, `Tenor Saxophone`, or `Percussion`).
2. **Single-Player Utility First (Personal Cloud Crates):** Users are not recruited as unpaid crowdsourcing labor for strangers. Instead, producers mark `start_time` and `end_time` boundaries during their

own active digging sessions to organize personal sample libraries, ensuring immediate single-player value so they never have to re-scrub a saved 5-minute video weeks later.

3. **Multi-Purpose Structural Reuse:** A single 5-minute composition typically contains multiple distinct musical passages (such as an unlayered electric guitar intro at `0:00--0:14` and an open drum break at `2:10--2:18`). Consequently, one producer's private bookmarking session creates persistent structural landmarks that other creators with entirely different sampling objectives can immediately reuse.

To formalize user interactions and guide subsequent system modeling, we define three canonical User Stories with testable Acceptance Criteria:

User Story 1 (Discovery): As an aspiring bedroom producer, I want to filter catalog recordings by localized objective preset tags (e.g., `[Solo Rhodes] + [No Drums]`) so that I can immediately discover unlayered harmonic progressions without scrubbing through full albums.

- *Acceptance Criteria 1:* Query results return timestamped passage cards rather than undifferentiated whole tracks.
- *Acceptance Criteria 2:* Selecting a result positions the embedded player at the segment cue point within external player tolerances ($\leq$ 1.5–2.0s keyframe drift).

User Story 2 (Auditioning): As a beatmaker, I want candidate passages to loop between defined temporal boundaries so that I can evaluate their sampling potential without manually restarting playback.

- *Acceptance Criteria 1:* The player returns to the start timestamp upon reaching the segment endpoint, functioning as a rough auditioning loop for evaluating musical potential (acknowledging $\approx$ 100–300 ms client buffer turnaround latency rather than sample-accurate metronomic looping).

User Story 3 (Curation): As an active crate-digger, I want to bookmark specific 4-to-8 bar passages and attach acoustic descriptors into my Personal Cloud Crate so that I can maintain an organized cloud reference library without consuming local drive storage.

- *Acceptance Criteria 1:* Bookmarked timestamps and tags persist in the user's private collection across sessions.

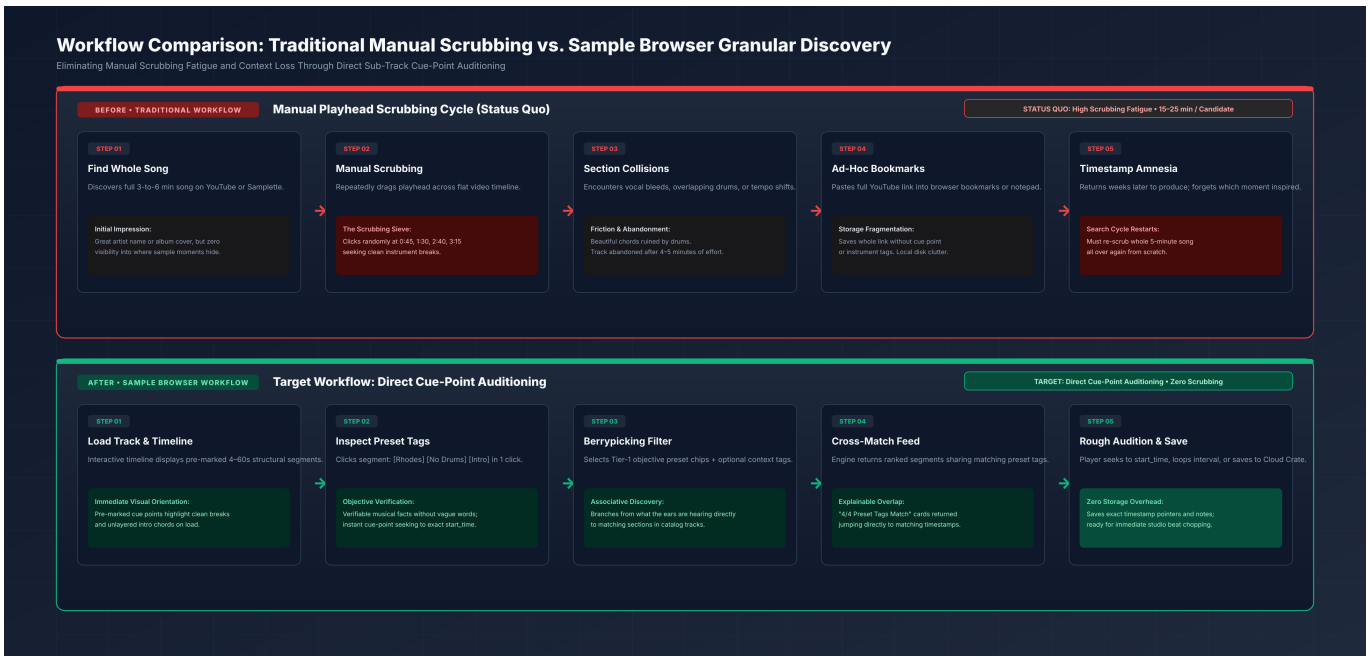


Figure 4 - Workflow Comparison: Traditional Manual Scrubbing vs. Sample Browser Granular Discovery.

4. Conceptual Interface Design

Figure 5 illustrates the conceptual user interface design of Sample Browser. The interface features the embedded track header, the segmented interactive timeline highlighting detected passage boundaries, an acoustic preset filter panel, and a synchronized passage feed.

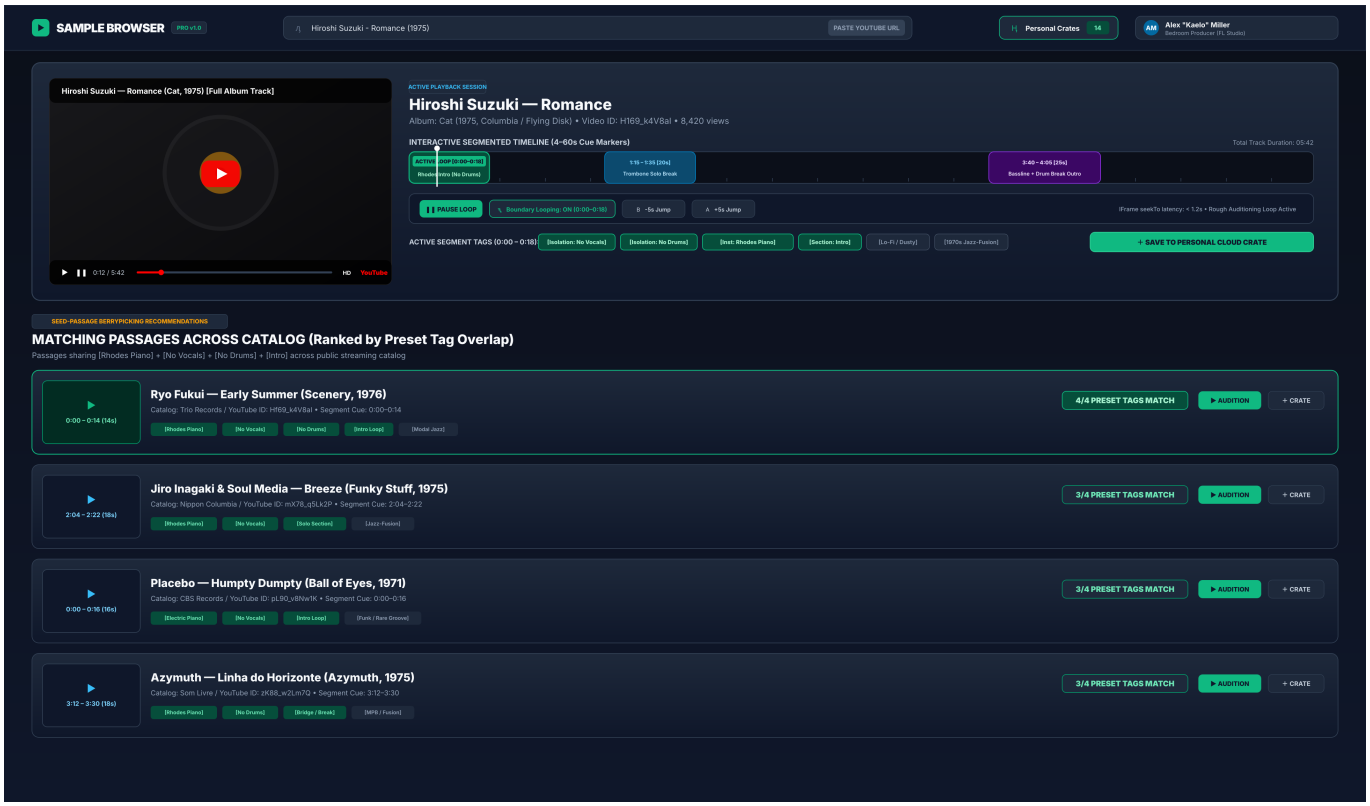


Figure 5 - Sample Browser Interface Wireframe: Embedded Player, Segmented Timeline, and Passage Curation Drawer.

The Internal Architecture Debate: Crowdsourcing vs. Automated MIR

During conceptual design, a critical architectural debate emerged between team members Leonid Tiscenco and Bogdan Glavan regarding data ingestion and catalog scalability:

- **The Automated MIR and Subjectivity Dilemma (Leonid):** Leonid argued that relying entirely on crowdsourced manual tagging is vulnerable to severe cognitive fatigue, inconsistent labeling, and malicious data entry. Furthermore, musical perception is fundamentally subjective—what one user tags as ”energetic soul,” another labels ”vintage pop,” as demonstrated by Flexer [50] in studies of low inter-rater agreement for perceptual music similarity, threatening database coherence. Simultaneously, Section III.E of the YouTube Developer Policies legally prohibits downloading audio streams for server-side automated MIR analysis, and producer gatekeeping (hiding rare samples from competitors) creates a severe cold-start risk for public crowdsourcing.
- **The Curatorial Incentive and Audio Context Model (Bogdan):** Bogdan countered that producers already spend hours digging and organizing personal sample notes. By coupling passage tagging with a personal cloud bookmarking utility, producers gain immediate private utility from bookmarking passages. Furthermore, producers frequently reveal sample sources on platforms like WhoSampled [32] because production originality stems from how a sample is chopped, pitched, and mixed rather than the source itself. By pairing objective preset categories with instant audio segment playback, any subjective ambiguity is instantly clarified by listening. Moreover, catalog cold-start can be effectively seeded without audio processing by ingesting text chapter markers from YouTube and instrument credits from Discogs.
- **The Architectural Synthesis and Contingency Pivot:** The team converged on the **Two-Tier Preset Taxonomy + Instant Audio Context** model over a pure pointer architecture. This completely eliminates illegal server-side audio downloads while preventing subjective tag anarchy. Furthermore, the team established a formal **contingency pivot trigger**: if upcoming field validation interviews reveal that producers refuse to share sample tags publicly due to gatekeeping, the team will execute an immediate, documented pivot to a **Personal Tagged Crate-Digging & Bookmarking Workspace** (differentiated from basic bookmarking sites like `chop-stop.com` and `sampledigging.online` by structured sub-track tagging that preserves why a section was saved).

Value Proposition

To ensure clear product-market alignment, we structured the solution against Alexander Osterwalder’s Value Proposition Canvas (Table 1). The framework explicitly contrasts independent producer workflows against the platform’s architectural interventions:

We summarize this value intervention into a single operational promise:

Table 1 - Value Proposition Canvas Matrix for Sample Browser.

CUSTOMER PROFILE (Music Producer)	VALUE MAP (Sample Browser)
Customer Jobs: • Crate-digging across streaming catalogs (YouTube/Samplette) for unique sample material. • Locating clean, unlayered 4–8 bar sections (breaks, intros, solos) suitable for chopping. • Organizing and revisiting candidate sample moments discovered during digging sessions.	**Products & Services:** • Web-based sub-track passage browser interfacing with YouTube IFrame API. • Two-Tier Tagging System (Objective Preset Structural/Instrument tags + Secondary descriptors). • Timestamped Personal Cloud Crates and seed-passage recommendation feed.
Pains: • Wasting time manually scrubbing through 5-minute tracks whose sections don't match search goals. • Losing track of exact timestamps and instruments inside bookmarked YouTube videos or playlists. • Sample fatigue and subscription costs of overused commercial loop packs (Splice).	**Pain Relievers:** • One-click playhead jumping (`seekTo`) and auto-looping of pre-marked 4–30s segments. • Preset `[No Drums]`, `[No Vocals]`, and `[Solo Instrument]` tags exposing clean sections instantly. • Cloud timestamp bookmarking that saves the exact interval and instrument notes without local disk clutter.
Gains: • Discovering obscure, authentic musical passages faster. • Smooth "berrypicking" navigation from one inspiring segment directly to similar segments in other songs. • Instant audio verification of whether a tagged passage fits the current beat idea.	**Gain Creators:** • Tag-intersection recommendations linking directly to sub-track cue points across songs. • Immediate audio context alongside tags to eliminate descriptive ambiguity. • Zero-cost web access over public streaming links without hosting or downloading files.

"For independent music producers who lose time scrubbing through full-length tracks on YouTube to find usable sample sections, Sample Browser is a passage-level discovery and bookmarking platform that connects creators directly to timestamped, auto-looped musical intervals categorized by objective structural and instrumental preset tags."

2.3 Target Group and Customer Validation

Preliminary Business Model Hypotheses

To ensure alignment with sustainable product design principles, Sample Browser defines a preliminary two-tier freemium model:

- **Free Community Tier:** Core web-based passage discovery, embedded player auditioning, public tagging, and personal cloud bookmarking.
- **Pro Creator Tier:** Advanced multi-tag intersection filtering, unlimited private Smart Crates, and DAW marker/timecode export (CUE/MIDI timecodes).

Target Group Segmentation

Based on demographic and workflow analysis, our target market is subdivided into three primary creator archetypes (Table 2). In the Tracklib creator survey, these three categories account for 93% of respondents (58% aspiring career producers, 25% hobbyists, and 10% full-time sync composers/sound designers). The remaining $\approx$ 7–8% of respondents in the survey represent absolute beginners who do not

yet use sample chopping or archival reference workflows, and are therefore excluded from our core target archetypes.

Table 2 - Target Group Segmentation Matrix for Sample Browser.

Segment	Market Share	Primary Production Goal	Primary Pain Point
Aspiring Career Producers	$\approx 58\%$ (Tracklib survey)	Releasing original beats, building client portfolios, seeking unique identity.	Limited time alongside day jobs; fatigue from manual digging.
Hobbyists & Beatmakers	$\approx 25\%$	Evening/weekend beat-making, community beat battles (lofi/boom-bap).	Frustration with repetitive commercial sample packs on Splice/Looperman.
Sync Composers & Sound Designers	$\approx 10\%$	Sourcing distinct vintage textures and acoustic references for media scoring.	Inability to search catalogs by specific timbral or solo instrument cues.

User Persona

To ground our architectural decisions in real creator workflows, we synthesized the archetypal persona of **Alex "Kaelo" Miller** (22, Bedroom Producer). Alex works full-time in technical support and produces hip-hop and neo-soul in his spare time (8–12 hours a week) using FL Studio and Ableton Live. Every weekend, he spends hours crate-digging through obscure 1970s Japanese jazz-fusion and Brazilian MPB recordings on YouTube.

His core friction captures the exact operational failure of current streaming search interfaces:

> *"I find an incredible 6-minute jazz-fusion album on YouTube. I know there is a 4-bar Rhodes chord progression somewhere in track 4, but I spend 15 minutes scrubbing the playhead back and forth trying to isolate it without drums over it. By the time I locate it, my inspiration is gone."*

Alex does not need another algorithmic playlist or generic radio feed. He wants to select an acoustic tag, jump straight to the unlayered passage, and instantly audition five other recordings sharing that exact sound.

Persona 2: Annie Lee (20, Student Sound Designer / Hobbyist Producer)

Annie is a media informatics student producing ambient soundscapes and lofi tracks. She works in sporadic weekend sessions and avoids commercial sample marketplaces due to repetitive, overused loops. Her primary friction is finding coherent instrumental textures across diverse world recordings. She needs a tool to rapidly audition isolated acoustic passages in context without breaking her creative immersion.

Customer Validation Hypotheses (Midterm 2 Roadmap)

In strict accordance with the PCAS course guideline rubric ("Define hypotheses about your solution's features and about your target groups"), all four validation hypotheses are explicitly partitioned into

target group/problem hypotheses and solution feature hypotheses. Each hypothesis is grounded in observable user behaviors and workflow friction so that it can be directly confirmed or refuted through qualitative interviews prior to software construction:

Category A: Target Group and Problem Hypotheses

- **H1 (Problem & Workflow Friction Hypothesis — Sub-Track Scrubbing & Isolation):** When hunting for samples in full-length recordings (e.g., on YouTube or Samplette), producers spend substantial time listening to and manually scrubbing through songs whose sections do not match their structural goals (such as finding a clean instrumental passage without vocals or heavy drums), frequently abandoning otherwise promising tracks. *Confirmed if* $\geq 60\%$ of interviewed sampling producers describe manual playhead scrubbing and the lack of clean/isolated sections as a recurring reason for slow digging sessions or abandoning candidate songs. *Refuted if* producers report that finding the right timestamp inside a song takes negligible effort, or that they simply grab any layered section and rely on DAW AI stem-separation without caring about clean sections.
- **H4 (Target Segment Hypothesis — Unlicensed Public Catalogs vs. Curated Libraries):** Small-scale and independent producers do not sample purely for commercial releases and actively prefer digging in unlicensed public full-track catalogs (YouTube, Samplette) over paid pre-curated libraries (Splice, Tracklib) due to cost, catalog variety, and commercial loop fatigue. *Confirmed if* independent/hobbyist respondents confirm they regularly sample from YouTube/public streams for practice, beat battles, or underground releases without being deterred by sample clearance. *Refuted if* respondents exclusively use royalty-free packs (Splice) and refuse to touch YouTube recordings due to copyright fear.

Category B: Solution Feature Hypotheses

- **H2 (Solution Feature Hypothesis — Two-Tier Preset Tags, Audio Context & Berrypicking Navigation):** Producers rely primarily on objective categories (instruments, section type/breaks, tempo) while distrusting standalone subjective mood/texture tags—and find it easier to discover samples by branching out from the tags of a specific audio segment they are currently hearing (*berrypicking*) rather than inventing text queries from scratch. *Confirmed if* producers prioritize instrument/structural filters, report that vague mood/vibe tags on existing platforms are unreliable without immediate audio preview, and naturally search by following associative chains of related tracks. *Refuted if* producers report that free-form text search or random browsing without tags is completely sufficient for their workflow.
- **H3 (Solution Feature Hypothesis — Timestamp Bookmarking Utility vs. Community Sharing / Gatekeeping):** Producers who save candidate songs online struggle to keep track of specific sub-track timestamps and reasons for saving them, and are willing to tag and share song sections using fast preset

categories as part of organizing their own sample library (rather than keeping all discovered sources strictly private). *Contingency Pivot Trigger:* If producers strongly want timestamp bookmarking and tagging for themselves but refuse to share tags publicly due to gatekeeping, the team will execute a planned soft pivot to a **Personal Crate-Digging & Bookmarking Workspace** (differentiated from basic bookmarking sites like `chop-stop.com` and `sampledigging.online` by structured sub-track tagging).

Customer Interview Protocol

For Midterm 2, the team has established a comprehensive 10-question semi-structured interview protocol grounded in *The Mom Test* methodology. In accordance with course validation guidelines, the team will conduct recorded video interviews first, targeting a sample size of $N \approx 20$ active music creators (4 interviews per team member), and back them up with a supplementary Google Forms quantitative survey to cross-validate findings across a broader pool of music creators while eliciting past operational behaviors rather than speculative opinions:

Block 1: Producer Background & Screening (Context)

1. **[BK]** *"How would you describe the main genres of music you make, and what inspires your production style?"*
2. **[BK + H4]** *"Do you currently release music under an alias, collaborate with artists, or make beats primarily for personal enjoyment, practice, or underground communities?"*
3. **[Screening]** *"When making a track, how much do you rely on sampling existing recordings versus composing from scratch with virtual instruments (VSTs) or hardware? Roughly how many samples did you use on your last sampled track, and why did you choose them?"*

Block 2: Unlicensed Song Digging vs. Pre-Curated Libraries (Testing H4)

4. **[H4]** *"Do you mostly use pre-curated royalty-free sample libraries (like Splice or Looperman), or do you prefer sampling directly from original full-length songs (on YouTube, Samplette, Tracklib, vinyl)?"*
    - *Follow-up 4a:* What makes you choose one over the other—is it convenience, cost, sound uniqueness/fatigue, or something else?
    - *Follow-up 4b:* Which free or paid tools/sites do you use most often to find samples? Has sample licensing/copyright ever been an issue for the music you make, or does it not affect your releases?

Block 3: Last Concrete Digging Session & Inside-Track Scrubbing (Testing H1)

5. **[H1]** *"Think about the last specific time you searched for a sample inside full-length songs online. What kind of sound or musical part were you looking for, and how long did you spend searching before you felt you spent enough time?"*

- *Follow-up 5a:* Roughly how many tracks did you click through and preview before you found one you actually used?

6. **[H1]** *"When you click on an unfamiliar 3-to-6 minute song on YouTube or Samplette, how do you actually scan through the track to find the exact 4-to-8 bar section you want?"*
    - *Follow-up 6a:* Have you recently found a song with a great sound or instrument, but ended up skipping it because the section wasn't usable? What ruined that section for you (e.g., vocals/-drums overlapping, tempo shifts)?
    - *Follow-up 6b:* Do you specifically hunt for clean, isolated sections in a song (like an intro, break, or solo), or do you use AI stem separation in your DAW on any part of the song? Why?

Block 4: Tag Reliability, Audio Context & Berrypicking (Testing H2)

7. **[H2]** *"What was the last sound you searched for using text keywords or category filters, and how did you describe it? How often do you find a good sample from a single text search query?"*
    - *Follow-up 7a:* When browsing platforms that use tags or filters (like Splice, Tracklib, Samplette, or RYM), which categories do you actually trust and rely on (e.g., instruments, BPM, song sections), and which tags do you avoid or find confusing/inaccurate (e.g., mood or texture words)?
8. **[H2]** *"Think of the last time you needed another sample to go with one you already had (or wanted to find more samples similar to one you just heard). How did you go about finding it?"*
    - *Probe (only if it doesn't come up):* Did anything about the first sample guide where or how you looked next?
    - *Follow-up 8a:* Think of the last time you picked a sound or a song because of how it was labelled—a mood, a genre, or an instrument. Did it match what you expected once you heard it?

Block 5: Personal Sample Storage, Timestamps & Community Contribution (Testing H3)

9. **[H3]** *"Do you keep a personal library, playlist, or folder where you save songs or samples you discovered for future projects? Do you store them locally on your drive or online (e.g., YouTube playlists, bookmarks, sites like chop-stop / sampledigging)?"*
    - *Follow-up 9a:* How do you categorize what you save? When you come back to a saved full-length song weeks later, how do you remember which exact timestamp had the sample and why you saved it?
10. **[H3]** *"Have you ever shared your sample finds, digging playlists, or song sources with other producers, or do you prefer to keep your found songs private?"*
    - *Follow-up 10a:* If you had a fast way to bookmark exact song sections (Start–End timestamps)

with preset instrument/section tags for your own online library while digging, would you leave those tagged sections public to help build a community catalog (with credit on your profile), or keep them private? Why?

- *Follow-up 10b (Gatekeeping probe):* Has there ever been a time when you found a great, rare sample and deliberately kept the song title secret until your beat was finished or released to prevent others from flipping it first?

2.4 Comparative Analysis and Strategic Evaluation

Benchmarking Against Existing Solutions

To evaluate how Sample Browser positions itself within the current creator landscape, it is benchmarked against representative tools across the sample discovery workflow:

- **Splice:** The commercial market leader offering pre-cut, isolated royalty-free loops and one-shots with sound-matching and DAW plugin integration [21, 29].
- **Samplette.io:** A web-based crate-digging interface applying metadata and geographical filters over YouTube's catalog to surface obscure recordings [25, 22].
- **Tracklib:** A commercial sampling marketplace offering full-length master recordings categorized by genre and BPM, with integrated licensing and bar-chopping tools [38, 24].
- **Sononym:** A desktop-based sample librarian that uses machine learning to organize and search local audio files based on acoustic similarity (timbre, pitch, brightness).
- **Cyanite.ai:** A B2B enterprise audio intelligence platform offering text-to-audio and audio-to-audio similarity search with segment-level scoring for sync agencies and publishers [5, 6].

Table 3 - Comparative Analysis: Sample Browser vs. Market Alternatives.

Platform	**Index Granularity**	**Query Paradigm**	**Catalog Type**	**Isolation Focus**	**Setup Friction / Cost**
Splice	Individual loop	Text / Sound Match	Curated proprietary	Pre-cut isolated loops	High (Commercial subscription)
Samplette.io	Whole track	Random / Discogs	Public (YouTube)	None (Full track)	Zero (Web-based)
Tracklib	Track + Bar loops	Metadata / Mood	Pre-cleared master (100k)	Partial (Stems on select tracks)	Low (Subscription + clearance)
Sononym	Local sample files	Audio similarity	Local user disk	N/A (Short files only)	High (Desktop install & license)
Cyanite.ai	Track & Interval	Text / Audio Reference	Enterprise B2B	Full mix (Segment scores)	High (B2B API contract)
Sample Browser (Ours)	**Passage (4–30s)**	**Two-tier preset & contextual tags**	**Public (YouTube)**	**Yes ([No Vocals] / [No Drums] / [Break] presets)**	**Zero (Web-based)**

SWOT Analysis

A comprehensive SWOT analysis evaluates the strategic positioning of Sample Browser within the creator tooling ecosystem:

Table 4 - SWOT Analysis of Sample Browser.

STRENGTHS (S)	**WEAKNESSES (W)**
• **Sub-Track Granularity:** Indexes at the 2–16 bar interval level, designed to reduce manual scrubbing friction (validation pending). • **Explainable Discovery:** Transparent acoustic descriptors rather than opaque black-box recommendations. • **Zero Hosting Overhead:** Uses compliant YouTube IFrame streaming, requiring no high-cost media storage. • **Contextual Playback:** Allows auditioning passages within their original musical arrangement.	• **Cold-Start Catalog Challenge:** Platform utility in early stages relies on pre-tagged passage volume, mitigated by automated text-based YouTube Chapter/description timestamp ingestion and Discogs instrument credit mapping. • **Third-Party Dependency:** Vulnerable to YouTube video takedowns, regional geo-blocks, or API embed policy changes. • **No Direct Audio Export:** Cannot offer one-click audio downloads under YouTube API terms; users must acquire source audio externally.
OPPORTUNITIES (O)	**THREATS (T)**
• **Rising Sample Pack Fatigue:** Growing producer backlash against sterile, overused commercial sample libraries (Splice) [21]. • **DAW Plugin Integration:** Potential to port the web interface into an embedded VST/AU plugin directly inside Ableton or FL Studio. • **Catalog Rights Monetization:** Future partnership pathways with archival labels to monetize dormant back-catalogs.	• **API Policy Lockdowns:** Risk of streaming providers restricting embedded player functionality or iframe event telemetry. • **Market Incumbent Expansion:** Established tools (e.g., Tracklib or Samplette) expanding natively into segment-level tagging. • **Hardware Keyframe Drift:** Hardware keyframe latency in embedded players (seekTo drift) and third-party streaming API policy modifications. • **Copyright Uncertainties:** Legal sensitivities surrounding sample clearance for archival recordings in commercial releases.

Analytical Synthesis

As demonstrated in Table 3 and Table 4, Sample Browser occupies a distinct, unoccupied niche at the intersection of streaming discovery and music production. While existing platforms force creators to choose between pre-cut, overused commercial loops (Splice) or tedious, unstructured digging across full-length tracks (Samplette, YouTube), Sample Browser addresses this divide by delivering granular passage-level indexing directly over the world's largest public music catalog.

3 Midterm 2 Roadmap: Planned Empirical Validation and Systems Modeling

The second project milestone will transition **Sample Browser** from conceptual problem analysis to empirical field validation and rigorous software engineering modeling. This chapter establishes our planned verification methodology, specifies operational system requirements, outlines the behavioral workflows to be modeled using unified modeling language (UML) artifacts, and maps the planned user experience architecture.

3.1 Customer Validation and Empirical Findings

3.1.1 Methodology and Participant Demographics

Empirical validation will prioritize qualitative depth by conducting recorded video interviews first, targeting a sample size of $N \approx 20$ independent music artists, beatmakers, and producers (4 interviews conducted per team member). To cross-validate findings and measure the statistical breadth of identified behaviors across a wider community of music makers, these qualitative video interviews will be backed up by a supplementary Google Forms quantitative survey.

Interviews may be conducted in any language (English, Romanian, Russian) provided that the participant is an active music maker, artist, or producer. When interviewing personal acquaintances or university peers, team members will adhere strictly to neutral, non-leading *The Mom Test* questioning principles to prevent confirmation bias. To ensure ecological validity, participants will be recruited across aspiring career beatmakers ($> 50\%$), hobbyist creators, and sound designers.

Participant recruitment will be structured across four distinct operational channels:

1. **Known Musician Contacts, Local University & Chișinău Artist Network (5–6 participants):** Fellow student producers at the Technical University of Moldova (UTM/FCIM), local Chișinău musicians, and personal acquaintances who create music (conducted via recorded video call or in person under strict non-biased protocol).
2. **Direct Instagram Outreach (5–6 participants):** Direct outreach (”shot in the dark” DMs) to independent beatmakers, underground music artists, and producers actively posting studio and sampling reels on Instagram.
3. **Dedicated Sampling and Beat-Battle Discord Communities (5–6 participants):** Active producers in Samplette / Crate-Digging Discord channels, Kenny Beats / Beat Battle communities, `r/makinghiphop` Discord, Lo-Fi Hip-Hop producer servers, and Stones Throw Beat Battles (STBB).
4. **Small YouTube / SoundCloud ”Sample Flip” Creators and Regional Telegram Chats (3–4 participants):** Independent producers posting sample-based beatmaking videos (”samplette flip”, ”youtube sample beat”) and active participants in regional Telegram producer communities who list public con-

tact handles.

Each interview will be executed by two team members (a Lead Interviewer and a Note-Taker/Scribe) following the 10-question protocol ($Q1$–$Q10$) grounded in *The Mom Test*. In strict accordance with the course guideline directive (*"Record videos and make photos of these interviews to refer to them when documenting your findings"*), verbal consent will be obtained to record remote sessions (via Google Meet / Zoom with screen shares) and log verbatim quotes. Post-interview responses will be synthesized into standardized 1-page coding cards evaluating hypothesis validation status.

3.1.2 Qualitative Interview Insights

Transcripts from qualitative customer development sessions will be coded thematically to evaluate hypotheses $H1$ through $H4$:

- **H1 (Scrubbing & Isolation Friction):** Evaluating whether manual scrubbing and the absence of clean intro/break sections represent the primary time sink during crate-digging, and whether modern DAW stem separation tools resolve or merely shift this friction.
- **H2 (Objective Preset Tags & Berrypicking):** Verifying whether producers prioritize objective structural/instrument tags over subjective mood labels, and whether browsing naturally follows associative seed-passage chains.
- **H3 (Curation & Contribution vs. Gatekeeping):** Assessing how producers organize discovered timestamps, and whether willingness to contribute to a community index is confirmed or triggers our contingency pivot to a **Personal Crate-Digging Workspace**.
- **H4 (Unlicensed Digging vs. Curated Libraries):** Confirming whether independent producers actively prefer digging across public full-track catalogs (YouTube, Samplette) due to cost and commercial loop fatigue on platforms like Splice.

3.1.3 Task-Completion Evaluation (Wizard-of-Oz Testing)

To validate interface ergonomics and task-completion velocity prior to full-scale software implementation, participants will execute identical audio retrieval tasks under two experimental conditions:

1. **Baseline Condition:** Unstructured crate-digging across public streaming catalogs (manual scrubbing and external note-taking).
2. **Prototype Condition (Wizard-of-Oz):** Retrieval using pre-indexed acoustic passage cards with direct timestamp seeking and automatic loop boundaries.

Metrics recorded will include time-to-first-usable-passage, session abandonment rate, and perceived cognitive friction measured via the NASA-TLX assessment framework. The test will explicitly evaluate interface flow and perceived value without misattributing findings to automated MIR algorithms.

3.2 System Requirements Specification

3.2.1 Functional Requirements

Based on creator workflows and the pointer-based YouTube IFrame architecture, the functional boundaries of Sample Browser comprise:

- **FR-01 (Two-Tier Tag Filtering):** The system shall allow users to filter passage candidates using primary preset objective tags (isolation, instrumentation, and section role) with optional secondary contextual descriptor chips.
- **FR-02 (Sub-Track Passage Retrieval):** The system shall return localized 4–30 second segment cards with associated timestamps and preset and contextual tags rather than entire compositions.
- **FR-03 (Embedded Cue Synchronization):** The player shall interface with the client-side YouTube IFrame API to position the playhead directly at the designated passage cue point (`seekTo`) upon card selection.
- **FR-04 (Automated Boundary Looping):** The embedded player shall continuously loop playback between the defined segment start and end timestamps until paused or redirected.
- **FR-05 (Personal Cloud Crates & Passage Submission):** Authenticated users shall be able to bookmark timestamped passages into private cloud crates and propose tags for community catalog inclusion.

3.2.2 Non-Functional Requirements

- **NFR-01 (Platform API Compliance):** The architecture shall strictly adhere to Section III.E of the YouTube Developer Policies by operating purely through embedded client-side playback, with zero raw audio stream downloading, scraping, or server-side media caching.
- **NFR-02 (Cue Seeking Latency):** Playhead positioning and buffer initialization shall execute within $\leq$ 1.5 seconds under standard broadband conditions, bounded by third-party iframe keyframe alignment.
- **NFR-03 (Backend API Response Time):** Metadata query latency for passage filtering and tag intersections shall not exceed 200 ms for 95% of requests.

3.3 Structural and Behavioral UML Modeling

3.3.1 UML Activity Diagrams

Activity models will formalize the sequential and concurrent logic of the system:

- **Discovery Flow:** Query submission $\rightarrow$ Tag intersection indexing $\rightarrow$ Candidate list rendering $\rightarrow$ User card selection $\rightarrow$ IFrame cue seeking $\rightarrow$ Loop listener loop activation.
- **Curation Pipeline:** User timestamp marking $\rightarrow$ Tag family assignment $\rightarrow$ Profanity and duplication filter $\rightarrow$ Persistence in PostgreSQL $\rightarrow$ Community appeal threshold verification.

3.3.2 UML Use Case Modeling

The operational interactions will model three primary actors:

- **Music Producer:** Searches passages by acoustic tags, audits contextual loops, and exports timecode references.
- **Community Contributor:** Marks candidate intervals, assigns acoustic tags, and manages personal crates.
- **System Moderator / Admin:** Audits reported erroneous tags, resolves taxonomic collisions, and monitors ingestion health.

3.3.3 UML Sequence Diagrams

Sequence diagrams will model asynchronous message exchanges across system boundaries, documenting the lifecycle of an acoustic search query, the communication handshake with the embedded player sandbox, and error recovery handling during stream unavailabilities.

3.4 Visual Interface Design and Interaction Architecture

3.4.1 Wireframes and Component Hierarchy

The interface architecture will follow a progressive disclosure layout designed for dark-mode studio environments. The component hierarchy centers the interactive segmented timeline, flanked by persistent acoustic filter drawers and a low-latency passage candidate drawer.

3.4.2 High-Fidelity Mockups

High-fidelity designs will reflect production-ready responsive layouts, detailing active playing states, tag selection badges, timeline segment highlights, and mobile companion views.

3.4.3 User Flow Diagrams

User flow networks will document navigational state transitions, delineating paths between initial landing, query refinement, continuous auditioning sessions, and private crate management.

4 Conclusions

The Midterm 1 milestone established the conceptual and analytical foundations of **Sample Browser**, resolving key product design questions through a Problem-Based Learning framework. By discarding our initial, unfeasible concept of an alternative consumer streaming shell, we focused on a genuine workflow bottleneck: independent producers losing hours scrubbing through long recordings to find isolated, sample-worthy passages.

Our domain research confirmed that contemporary streaming platforms index music strictly by whole-track metadata. This leaves creators trapped between generic, overused loop libraries like Splice and tedious, unstructured crate-digging across YouTube or Samplette. **Sample Browser** bridges this divide. By dividing tracks into 4–30 second passages, assigning structured acoustic tags, and using the YouTube IFrame API for compliant playback seeking and looping, our architecture protects the producer's creative focus.

This milestone yielded four tangible deliverables for our engineering roadmap:

- **A validated problem definition** supported by academic MIR literature and empirical creator surveys (Tracklib, UK Musicians' Census).
- **A candid retrospective of project failures**, recording our team's pivot away from vocal minority feedback, passive user habits, and closed API barriers.
- **Four falsifiable validation hypotheses (H1–H4)** paired with a semi-structured interview protocol for field testing.
- **A competitive and SWOT benchmark** clearly positioning our application against Splice, Samplette, Tracklib, Sononym, and Cyanite.

With the domain analysis and system concept established, our team is prepared for Midterm 2. The next milestone will focus on validating our hypotheses through $N \approx 20$ recorded video interviews with active music creators (4 per team member) backed by a supplementary quantitative survey, formalizing software requirements through UML modeling (Activity, Use Case, and Sequence Diagrams), and building interactive interface wireframes.

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